

Application of Random Forest and SHAP for real-time Taxi drivers' traffic violations detection and feature analysis

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Abstract: To effectively explore the impacts of several key factors on taxi drivers' traffic violations and provide traffic management departments with scientific decisions to reduce traffic fatalities and injuries. 43,458 electronic enforcement data on taxi drivers' traffic violations in Nanchang City, Jiangxi Province, China, from July 1, 2020, to June 30, 2021, were utilized to explore the characteristics of traffic violations. Eleven actors affecting traffic violations, including time, road conditions, environment, and taxi company were analyzed. Firstly, the ensemble method Balanced Bagging Classifier (BBC) was applied to balance the dataset. The results showed that the original imbalanced dataset's imbalance ratio (IR) reduced from 6.61% to 2.60%. Moreover, the method of Random Forest was adopted to establish the severity of taxi drivers' traffic violations prediction model, and the results showed that accuracy, m_F1, m_G-mean, m_AUC, and m_AP obtained 0.877, 0.849, 0.599, 0.976, and 0.957, respectively. Compared with the algorithms of Decision Tree, XG Boost, Ada Boost, and Neural Network, the performance measures of the prediction model based on Random Forest were the best. Finally, the Shapley additive explanation (SHAP) framework was used to improve the interpretability of the model and identify important factors affecting taxi drivers' traffic violations. The results showed that functional districts, location of the violation, and road grade were found to have a high impact on the probability of traffic violations; their mean SHAP values were 0.39, 0.36, and 0.26, respectively.

Keywords: Taxi drivers' traffic violations; impact factors; imbalanced dataset; Random Forest; SHAP

1. Introduction

Road traffic injuries have always been a critical issue for global health. The Global Plan Decade of Action for Road Safety 2021-2030 on the global status of road safety released in 2021 highlighted that 1.30 million road traffic deaths had been reported each year, and an additional 50 million people are suffering life-threatening injuries from motor vehicle-related crashes (Organization & Commissions, 2021). In China, although the mileage of urban roads only accounts for about 7.5% of the total mileage of all roads, the number of traffic accidents on urban roads accounts for 47% of the national total (Qu, Lin, Li, & Zhang, 2019). Taxi drivers' traffic violations are one of the leading causes of traffic accidents (Zhang et al., 2013). Studies have found that taxi drivers are prone to traffic violations driving, such as speeding, redlight running, and violating traffic markings, which quickly affect urban road traffic conditions and road safety (Tseng, 2013). By predicting the probability of the severity of taxi drivers' traffic violations and analyzing the impact factors of traffic violations to formulate targeted traffic safety measures, the accident rate will be effectively reduced.

The current sources of traffic violation data are mainly extracted from questionnaire surveys, self-reports, and case data reports of specific incidents (Atombo, Wu, Zhong, & Zhang, 2016). Data of this kind carry with them various well known limitations, such as subjectivity, incompleteness, and vagueness. Compared to the above data sources, the electronic enforcement tool could automatically detect traffic violations all day long and accumulate a large number of historical violation records, which is helpful for researchers in comprehensively and objectively analyzing the traffic violations of motor vehicles (Li, Abdel-Aty, Yuan, Cheng, & Lu,

2020). In this study, 43,458 electronic enforcement data of taxi drivers' traffic violations from Nanchang, Jiangxi Province, China, from July 1, 2020, to June 30, 2021, are used and employed by Web crawler technology to obtain other valid data. These violation data record the time, space, license plate, taxi company, code, and other information of traffic violations.

2. Literature review

2.1. Impact Factors of Traffic Violations

Numerous studies have shown that the four major factors of people, vehicles, roads, and the environment are the main risk factors affecting traffic violations (Das & Sun, 2016). The driver's gender, age, psychological state, and driving experience are the factors that affect the driver's traffic violations (Pantangi et al., 2020). For example, González-Iglesias, Gómez-Fraguela, and Luengo-Martín (2012) found that male drivers tend to drive riskier, such as speeding and drunk driving, while female drivers were more cautious than male drivers. Oppenheim, Oron-Gilad, Parmet, and Shinar (2016) demonstrated that younger drivers (under 34) were more prone to speeding. Moreover, studies have found that drivers with only one year of driving experience were inclined to commit traffic violations (Lucidi et al., 2010).

Drivers driving vehicles with different features and sizes exhibit different traffic violations. For instance, Inada, Ashraf, and Campbell (2021) evaluated the traffic violations in passenger vehicles, cargo vehicles, and motorcycles. The research showed that 57% of traffic violation vehicles were passenger vehicles. de Carvalho Ponce, Muñoz, Andreuccetti, de Carvalho, and Leyton (2011) found that private car drivers were more tempted to drunk driving than truck drivers in São Paulo. Due to the different nature of operation and driving experience, traffic violations such as speeding, driving not according

to the set driveway, random U-turns, and reversing were relatively common among taxi drivers(Li, Jia, & Shao, 2016).

Road type, road length, and the number of lanes affect motor vehicle traffic violations(Liu et al., 2020). For example, Hosseini, Mahdavi, and Nooghabi (2018) applied Poisson and the zero Truncated Poisson model to study traffic violations under different road conditions on Iranian highways. The results showed that adding one interchange or a passing lane reduced hourly violations by 5% and 64%, respectively. Turnpike increased the number of traffic violations by 238% compared to the highway. Huang, Sun, and Tang (2018) studied the relationship between the length of the road section and speeding and found that when the length of the road section is 1.0-1.5 kilometers, drivers were prone to speeding.

Environmental factors such as weather, temperature, visibility, and traffic law enforcement also affect motor vehicle traffic violations. For example, Zhang, Tan, and Jou (2016) found that rainy days and air pollutants reduced visibility, making drivers unable to observe road conditions and traffic control devices promptly. Walter and Studdert (2015) demonstrated that automatic camera enforcement using fixed or mobile cameras could effectively reduce traffic violations such as using mobile phones, speeding, drunk driving, and others.

2.2. Traffic Violations Prediction Analysis

Most previous literature focuses on using statistical analysis and machine learning methods for traffic violation prediction(Pan et al., 2020). Statistical analysis methods generally include Univariate analysis, Pearson correlation analysis, and other methods. For example, Akaateba, Amoh-Gyimah, and Amponsah (2015) used oneway analysis of variance (ANOVA) and

Bonferroni post hoc test analysis to investigate average traffic violations among drivers with further education and driving experience. The findings showed that the average frequency of traffic violations increased with driver experience, while the frequency of traffic violations for speeding and overtaking while prohibited decreases with educational attainment. Hong, Park, Lee, and Park (2019) analyzed the relationship between drivers, vehicle characteristics, traffic flow, and traffic violations through Pearson and partial correlation, and the results showed that traffic violations were identified to be influenced by the driver's physical condition, as well as vehicle characteristics.

In addition, machine learning methods generally include the Logit model, Decision Tree, Naive Bayesian Network, and other methods. For example, Alver, Demirel, and Mutlu (2014) employed a binary logit model to study the traffic violation characteristics of young drivers (18-29 years old). The results indicated that younger were less inclined to use seat belts. Moreover, the presence of other passengers in the car or showing off are two essential factors that led to speeding. Cheng, Lu, Zu, and Li (2019) applied the decision tree method to predict the types of speeding violations. The study found that many speeding violations occurred in high rainfall environments.

Previous studies have found that statistical analysis methods have high requirements for the integrity and accuracy of historical statistical data. The results are unsatisfying when using machine learning methods to process data samples with many categories(Kotsiantis, Zaharakis, & Pintelas, 2006). Compared with machine learning algorithms, the ensemble learning algorithm has better stability and can solve the endogenous variable relationship and multicollinearity between the various impact factors(Sagi & Rokach,

2018). In addition, the ensemble algorithm accuracy rate is higher than statistical learning models. For example, Wang, Liu, Xu, and Lv (2019) utilized a gradient boosting decision tree (GBDT) to study the impact factors of driver traffic violations, and the result showed that GBDT was the best model choice across all periods, with an acceptable average precision (AP) of 0.68. Xu, Bao, and Pradhan (2021) applied random forest algorithms to predict the driver's reaction. The accuracy rate reached 74.7 %, which effectively analyzed the impact of drivers, vehicles, and environmental factors on the driver.

Many scholars have studied the impact factors and the prediction methods of traffic violations. However, previous studies had the following limitations: Firstly, the data sources of traffic violations were mainly questionnaire surveys or case data reports of traffic violation cases, which could not comprehensively and objectively record information. Secondly, traditional statistical analysis methods and machine learning could not fit driving violations well. At the same time, previous studies had ignored the problem of unbalanced data sets, which made the overall classification performance of the model low. Finally, although some studies employ ensemble learning methods, they could not explain the output results well. Thus, the data of electronic enforcement and the data crawled from the Internet were exploited to research the taxi drivers' traffic violations. Furthermore, Balanced Bag Classifier (BBC) was employed to preprocess the data to obtain a balanced dataset. Secondly, a Random Forest model was utilized to construct a taxi driver's traffic violation severity prediction model. Finally, the Shapley additive explanation (SHAP) framework was adopted to improve the interpretability of the model and identify important factors affecting traffic violations.

3. Methodology

3.1 Balanced Bagging Classifier

Many methods have been adopted to address the data imbalance problem. They may be categorized into data level and algorithm level ones. Within the algorithm level approaches, ensemble algorithms are quite widely applied. Experimental comparisons have shown that bagging has advantages over boosting model generalizations and extensions (Galar, Fernandez, Barrenechea, Bustince, & Herrera, 2011). For example, Lango and Stefanowski (2015) demonstrated that the combination of bagging ensemble and under-sampling technique improved the classification of imbalanced data better than boosting and other ensemble methods. The balanced bagging classifier (BBC) formulation is divided into the following main steps: Firstly, dividing the imbalanced dataset into several subsets. Secondly, the under-sampling technique and the classifying estimators are applied to each subset. They are finally synthesizing the result of each subset. The flow and principles of the Balanced Bagging Classifier (BBC) are shown in Figure 1.

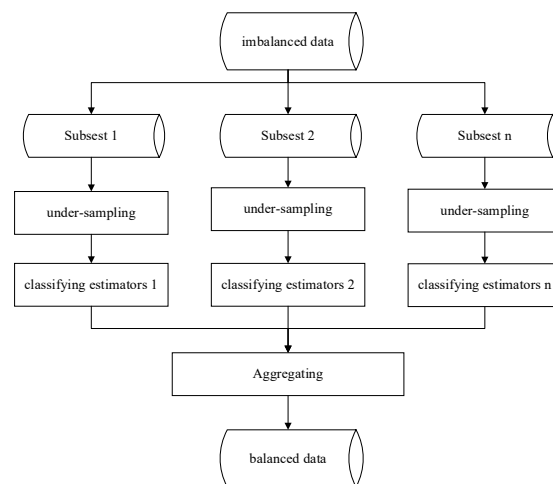


Fig. 1. Flow chart of the Balanced Bagging Classifier.

3.2 Random Forest (RF) algorithm

The RF algorithm trains multiple sample subsets, effectively enhancing the differences between the various classification models. Multiple sample subsets can improve the generalization and prediction ability of the algorithm. Compared with a single decision tree, the random forest model can handle data samples with outliers and missing values better, so it has stronger robustness (Dogru & Subasi, 2018).

Assuming that the original data set D contains n data bars and the sample feature dimension is M , K decision trees are established. The random forest algorithm is mainly described: from the original data set D , sampling with replacement is adopted k times to form K new sub-training sets. Randomly choose m features, where $m < M$. Using the new training set K and m features, learn k complete decision trees and get a Random Forest. The flow and principles of the Random Forest algorithm are shown in Figure 2.

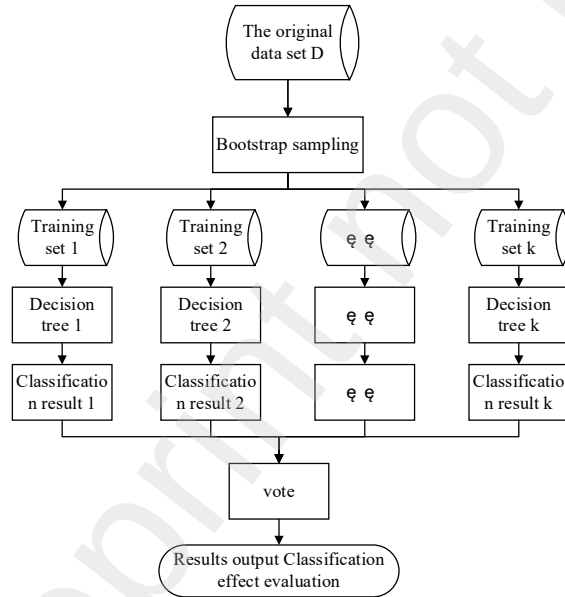


Fig. 2. Flow chart of the Random Forest algorithm.

Many hyperparameters of the random forest model affect the model's performance.

Hyperparameter optimization can tune the model performance. The most critical parameters in the Random Forest model are $n_estimators$, $max_features$, max_depth , and $min_samples_split$. In this study, the GridSearchCV algorithm and 10-fold cross-validation are applied to optimize hyperparameters of the Random Forest. The GridSearchCV function utilizes the grid search concept to search all parameter combinations in the parameter space to determine the optimal hyperparameters (Gan et al., 2021). Specifically, k -fold cross-validation is performed to search for the best combination of hyperparameters according to the specified range based on the n_MSE score.

3.3 Performance Measures

Accuracy, $macro_F1$ -measure (m_F1), $multiclass_geometric$ mean (m_G -mean), $micro_area$ under the receiver operating characteristic curve (m_AUC), and $micro_area$ under the precision-recall curve (m_AP) are commonly selected to evaluate model performance. The TP is the number of true positives at the leaf, FN is the number of false negatives at the leaf, FP is the number of false positives at the leaf, and TN is the number of true negatives at the leaf. The accuracy, precision, and recall measures are defined as follows:

$$Accuracy = \frac{TP + TN}{(TP + TN + FP + FN)} \quad (2)$$

$$Precision = \frac{TP}{TP + FP} \quad (3)$$

$$Recall = \frac{TP}{TP + FN} \quad (4)$$

$Macro_F1$ -measure considers both $macro_precision$ and $macro_recall$, and the $macro_F1$ -measure is defined as follows:

$$macro_P = \frac{1}{n} \sum_{i=1}^n P_i \quad (5)$$

$$macro_R = \frac{1}{n} \sum_{i=1}^n R_i \quad (6)$$

$$macro_F_1 = \frac{2 \times macro_P \times macro_R}{macro_P + macro_R} \quad (7)$$

The multiclass G-mean is defined as follows: m_G -mean represents the geometric average of the recall in each class. If any class receives a very low recall, the m_G -mean value will be taken down(Wang & Yao, 2012). This metric can evaluate the classification performance of the classifier among different classes. The calculation formula is as follows, where M is the number of categories:

$$m_G - mean = (\prod_{i=1}^M R_i)^{\frac{1}{M}} \quad (8)$$

The area under the receiver operating characteristic curve (m_AUC) and the area under the precision-recall curve (m_AP) evaluate the model performance. The precision-recall (PR) curve is better suited for comparing the performance of individual classifiers than the m_AUC . The advantages of m_AP arise because it does not incorporate correctly predicted absences and is, therefore, less prone to exaggerate model performance for unbalanced datasets(Khan & Rana, 2019).

4. Data collection and proccession

4.1. Data Collection

The traffic violation data used electronic enforcement data on taxi drivers' traffic violations collected in Nanchang, Jiangxi Province, China. The data has 43,622 records extracted from the electronic police system from July 1, 2020, to June

30, 2021, including time, space, license plate, taxi company, code, and other information on traffic violations.

To obtain additional information on traffic violation electronic enforcement data, we obtained daily rainfall, weather, and temperature from the National Oceanic and Atmospheric Administration (NOAA) National Center for Environmental Information (NCEI) through crawling technology. Furthermore, the nature of 32 taxi companies was extracted from the website <https://www.tianyancha.com/.html>.

To obtain the spatial location of the traffic violation data, we employed geocoding in the API of the AutoNavi Map service. The functional districts, roads, intersections, surrounding points of interest (POI), and the road grade were obtained from the address location field. After eliminating 164 data that could not be parsed to obtain accurate addresses, we obtained 43,458 accurate records.

4.2 Data Proccession

This study's severity of taxi drivers' traffic violations was divided according to the driving record points. Traffic violation deduction grades are divided into six types, namely 0, 1, 2, 3, 6, and 12. In this study, deductions of 0 and 1 were classified as slight infractions, deductions of 2 and 3 were classified as medium infractions, and deductions of 6 and 12 points were classified as severe violations. The specific descriptions of traffic violations can be obtained according to the traffic violation codes. The standard taxi drivers' traffic violations are shown in Table 1.

Table 1
Major traffic violations of taxis driver.

Traffic violation level	deducted points	traffic violation codes	Description of traffic violations	Frequency
slight	0	1039	Illegal parking	9,825
infractions	1	3019	Failure to use safety belt as required	4,015

		1345	Violating traffic markings	3,349
		1344	Violation of prohibition signs	1,141
	2	1244	Failure to use the safety belt as required when driving on the urban expressway	118
medium infractions		6023	Failure to follow the specified lane when driving on roads other than expressways and urban expressways	7,254
	3	1208	Failure to drive into the guide lane in the required direction of travel	3,609
		1357	Failure to yield to pedestrians	3,289
severe infractions	6	1625	Violating traffic signals	2,889
		1636	Speeding by 20% but less than 50%	352

As shown in Table 1, slight traffic violations mainly included illegal parking, failure to use safety belts as required, violating traffic markings, and violation of prohibition signs. Medium infractions traffic violations mainly included failure to use the safety belt when driving on the urban expressway, failure to follow the specified lane when driving on roads other than expressways and urban expressways, and failure to drive into the guide lane in the required direction of travel. Severe infractions traffic violations mainly

included violating traffic signals and speeding.

43,458 electronic enforcement data were applied in this study. The dataset included four factors related to taxi drivers' traffic violations: time, road conditions, environment, and taxi companies. The dependent variable was the level of traffic violations, and the remaining 11 characteristic variables were the independent variable. Descriptions of variables are presented in Table 2.

Table 2
Variable descriptions.

Type	Symbol	Variable name	Description	Frequency	
				N	(%)
Traffic Violations	Y	Traffic Violation Level	slight infraction (0)	21,524	49.53%
			medium infractions (1)	18,678	42.98%
			severe infractions (2)	3,256	7.49%
Time	x ₁	Time Periods	low frequency (0)	3,152	7.25%
			medium frequency (1)	11,151	25.66%
			high frequency (2)	29,155	67.09%
	x ₂	Seasons	spring (0)	11,956	27.51%
			summer (1)	10,506	24.18%
			autumn (2)	10,146	23.35%
			winter (3)	10,850	24.97%
		Weeks	weekday (0)	31,637	72.80%

Road Conditions	x_3		weekends (1)	11,821	27.20%
	x_4	Location of Violation	road (0)	17,920	41.24%
			intersection (1)	13,791	31.73%
			POI (2)	11,747	27.03%
			central business district (0)	12,247	28.18%
			The old town (1)	7,233	16.64%
	x_5	Functional Districts	external transportation area (2)	9,329	21.47%
			industrial area (3)	2,300	5.29%
			development zone (4)	6,082	14%
			municipally administered county (5)	6,267	14.42%
			national and provincial highways (0)	614	1.41%
			expressways (1)	4,596	10.58%
			primary (2)	10,819	24.90%
	x_6	Road Grade	secondary (3)	15,981	36.77%
			tertiary (4)	11,448	26.34%
Environment	x_7	Road Surface Conditions	dry (0)	23,315	53.65%
			damp (1)	13,339	30.69%
			wet (2)	6,804	15.66%
			sunny (0)	7,054	16.23%
	x_8	Weather	cloudy (1)	8,245	18.97%
			overcast (2)	13,558	31.20%
			rainy (3)	14,601	33.60%
	x_9	Environmental comfort	cool (0)	12,009	27.63%
			comfortable (1)	18,014	41.45%
			warm (2)	13,435	30.91%
	x_{10}	Visibility	poor visibility (0)	2,252	5.18%
			fair visibility (1)	11,426	26.29%
			good visibility (2)	29,780	68.53%
Taxi Company	x_{11}	Taxi Company Attributes	state-owned		
			corporation/group (0)	17,234	39.66%
			private company (1)	26,224	60.34%

5 Results

5.1 The Results of The Balanced Bagging Classifier

The results are highly biased towards the

majority class when using ensemble algorithms or machine learning models on imbalanced datasets. The Balanced Bagging Classifier (BBC) combines data level (under-sampling) and algorithmic level

(bagging), which is a method in Python's imbalanced learning toolbox. The dataset's characteristics before and after processing are shown in Table 3. Size is the number of samples, Class_0 is a slight infraction, Class_1 is a medium infraction, Class_2 is a severe infraction, and Min_Size is the number of minority samples. The

Imbalance Ratio (IR) is defined as the number of the majority class instances divided by the number of the minority class instances (Kotsiantis, Kanellopoulos, & Pintelas, 2006), which describes the degree of imbalance in a data set. In general, when the imbalance ratio is greater than 4, it will affect the classifier's performance.

Table 3

Description of imbalanced data preprocessing.

Dataset	Size	Class_0	Class_1	Class_2	Min_Size	IR (%)
Imbalanced data	43,458	21,524	18,678	3,256	3,256	6.61
BBC data	43,458	19,031	17,019	7,407	7,407	2.60

Table 3 shows that BBC effectively reduces the IR of the dataset. The IR of the original imbalanced data is 6.61%, and the preprocessed IR is 2.60%. Moreover, the data distribution characteristics processed by the ensemble method are consistent with the original unbalanced data, indicating the reliability of the preprocessed data.

5.2 The Results of The Random Forest Model

This study adopted the RF model to predict taxi drivers' traffic violations by involving multiple factors such as traffic violation time, road conditions, environment, and taxi companies. The experiments were set up using the scikit-learn library in Python. The dataset was divided into a training set (70%) and a test set (30%) adopted for model training and classification.

The model was optimized using 10-fold cross-validation and hyperparameter tuning. The steps of hyperparameter tuning: Firstly, drew the learning curve of each parameter and determined the range of best-performing parameters according to the trend and score of the learning curve. Furthermore, n_MSE scores searched the parameter set of hyperparameters during 10-fold cross-validation. The experiments were performed using the GridSearchCV function in python scikit-learns for hyperparameter tuning. After tuning, the

$n_estimators$, the max_depth , the max_depth , and the $min_samples_leaf$ set to 310, 4, 17, and 2, respectively. The performance results of the Random Forest model are shown in Figure 3.

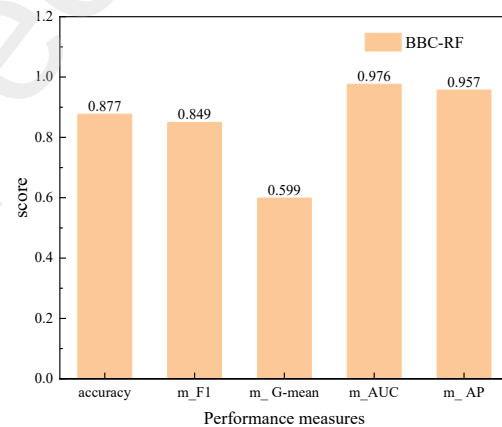
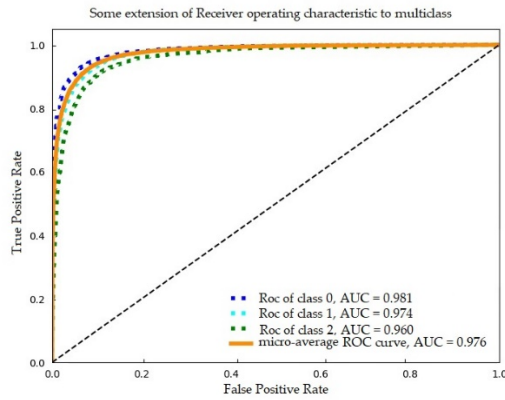
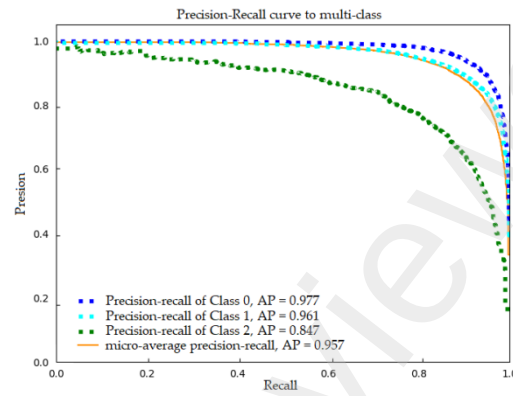


Fig. 3. Performance Results of BBC-RF.

Figure 3 shows the five performance measures of BBC-RF, including accuracy, m_F1 , m_G-mean , m_AUC , and m_AP . The accuracy and m_AUC values are 0.877 and 0.976, respectively, indicating that BBC-RF can predict taxi driver traffic violations well. In addition, the values of m_F1 and m_AP are 0.849 and 0.957, respectively, indicating that the BBC can solve the class imbalance problem well and effectively improve the model's fit.



(a) ROC curves



(b) Precision-Recall curves

Fig. 4. ROC curves and Precision-Recall curves of BBC-RF.

Moreover, Figure 4 shows the ROC curves and Precision-Recall curves of BBC-RF models. Figure 4(a) shows that the horizontal represents the false positive rate, the ordinate represents the true positive rate, and the area under the curve represents the AUC. Figure 4(a) shows that all ROC curves are higher than 0.9. The closer the ROC curve is to the upper left corner, the better the model's classification performance. Figure 4(b) shows that the horizontal represents the precision, the ordinate represents the recall, and the area under the curve represents the AP. The advantage of the precision-recall curve is that it is more sensitive to considering the impact of negative examples on the algorithm's performance, revealing the differences between algorithms that are not obvious in the ROC curve. The AP values of Class_0 and Class_1 are greater than 0.9. The AP value of Class_2 using BBC processed data is 0.847. From the ROC curve and precision-recall curve of each classifier, it can be seen that the BBC-RF method performs relatively well in predicting the taxi drivers' traffic violations.

The random forest model was tested using the original disequilibrium data, and the BBC processed data to verify the effect of different IR on the model results. The results of the Performance measures of the Random Forest

classifier on the original imbalanced data and the BBC data are shown in Figure 5. When the imbalance ratio is from 6.61% to 2.60%, the balanced bagging classifier (BBC) improved the results in terms of accuracy, m_F1, and m_G-mean. Figure 5 reveals that there has been a gradual increase in the number of m_G-mean. After using BBC to process the original imbalanced data, the m_G-mean increases from 0.256 to 0.599, indicating that the BBC effectively solved the problem of imbalanced data classification.

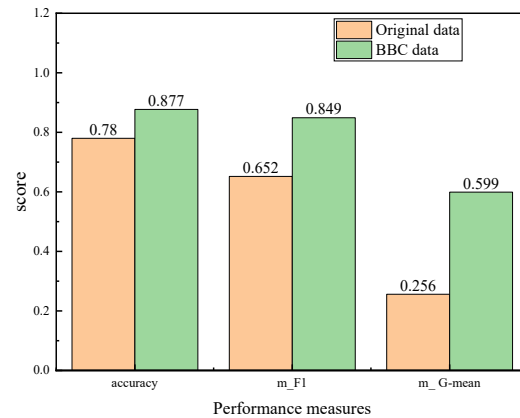


Fig. 5. Performance Evaluation of original imbalanced data and BBC data.

To verify the superiority of the random forest method, the Decision Tree, XG Boost, Ada Boost,

and Neural Network methods were utilized to analyze the data processed by BBC. the performance comparison of different methods is shown in Figure 6. Figure 6 shows that the accuracy, m_F1, m_G-mean, m_AUC, and m_AP of the Random Forest method are more prominent than other methods by contrast. The accuracy, m_F1, m_G-mean, m_AUC, and m_AP of the XG Boost method are slightly lower than the Random Forest method but higher than the other method.

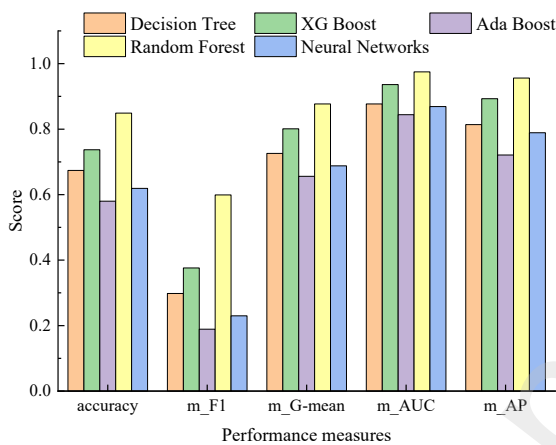


Fig. 6. Performance Evaluation of Various Methods.5.3 The Analysis of The Impact Factors

SHAP (Shapley Additive exPlanation) is an explanatory framework for explaining machine learning models. Specifically, SHAP adopts game theory concepts to identify the contribution of each input feature(Guo, Fu, & Sollazzo, 2021). In traffic safety, Parsa, Movahedi, Taghipour, Derrible, and Mohammadian (2020) used XG Boost to predict traffic accidents and applied SHAP to analyze the impact of individual features on the results. The trained Random Forest model was explained based on the SHAP framework to explore the influence of various factors on the

severity of taxi drivers' traffic violations.

Figure 7 is a ranking diagram of SHAP feature importance. The size of the mean SHAP value of each impact factor represents its impact on the severity of serious traffic violations. The larger the mean SHAP value, the greater the degree of influence.

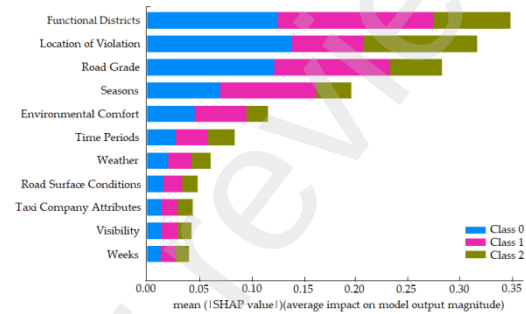


Fig. 7. Ranking diagram of SHAP feature importance.

As reflected in Figure 7, features such as “Functional Districts”, “Location of Violation”, “Road Grade”, “Seasons”, and “Environmental comfort” appear to be the top five influential features in taxi drivers' traffic violations. The mean SHAP values of functional districts, location of the violation, and road grade are 0.39, 0.36, and 0.26, respectively. We can see that the top five feature factors belong to the categories of "road condition," "time," and "environment," indicating that the combination of BBC-RF and SHAP can explain the features of most description categories.

Figure 8 shows scatter density plots of the SHAP value of each influence factor adopted. In this case, the feature importance of the model is proportional to the size of each factor’s absolute mean SHAP value. A feature with a positive SHAP value means that feature would make the predicted value increase, and a negative SHAP indicates a contrary effect.

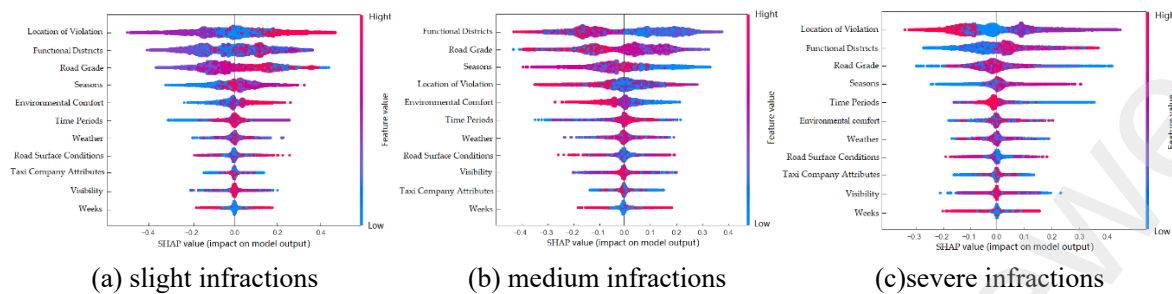


Fig. 8. SHAP Summary plot of input features.

From Figure 8(a), which shows the scatter density plots of SHAP values for slight violations, it can be seen that the location of traffic violations is the most crucial impact factor. There are three colors corresponding to the location of traffic violations, purple represents intersections, blue represents the road, and red represents a surrounding point of interest (POI). POI mainly includes residential, commercial, education, hospital, and entertainment. The SHAP value corresponding to red is more extensive, indicating that slight traffic violations can occur in places with points of interest near roads. Many studies have shown a direct relationship between the number and location distribution of the POI with traffic violations (Koochpayma, Tahooni, Jelokhani-Niaraki, & Jokar Arsanjani, 2019). There are three colors corresponding to the road grade, blue represents national and provincial highways, expressways and highways, purple represents the primary, dark purple represents secondary, and red represents tertiary. The SHAP value corresponding to blue and red is greater than zero, indicating that national and provincial highways, expressways and highways, and tertiary roads positively impact slight infractions. Since the city usually allows parking or temporary parking on roads that are not heavily trafficked and there are few parking violation signs on tertiary roads, taxi drivers are inclined to have parking violations.

From Figure 8(b), which shows the scatter density plots of SHAP values for medium

infractions, it can be seen that the functional districts of traffic violations are the most crucial impact factor. Purple represents the external transportation and industrial areas in the functional districts where traffic violations are located—corresponding to a SHAP value greater than zero, indicating that moderate violations will probably occur in the external transportation and industrial areas. The reason may be that the external transportation area is located at the core node of the central highway hub, with six exits of the city's outer ring highway, which have a complex traffic flow. The dark purple and purple in the road grade of traffic violation represent the primary and secondary roads, respectively. The SHAP value is greater than zero, indicating that medium infractions frequently occur on primary and secondary roads. A left-turn lane is generally set on the main road. It is not easy to meet the needs when the traffic flow is more traffic. If the queue length is too saturated, it may result in taxi drivers' failure to drive into the guide lane in the required direction of travel. There are more residential areas along secondary roads with more crosswalks, resulting in passing crosswalks without stopping to yield (Papadimitriou, Lassarre, & Yannis, 2016).

From Figure 8(c), which shows the scatter density plots of SHAP values for severe infractions, it can be seen that the location of the violation is the most crucial impact factor. The purple location of the violation represents the road intersection, indicating that severe infractions are tempted to occur at the road intersection.

Generally, most traffic signals and traffic markings are placed at the intersection. The probability of violating traffic signals is higher in the intersection(Fu, Pei, Wu, & Qi, 2013). Red represents development zones and municipally administered county, whose corresponding SHAP values are greater than zero, and taxi drivers often commit serious traffic violations. The reason may be that the development zone is distributed with many expressways and airport expressways, which are prone to speeding.

6. Discussion and Conclusions

The electronic enforcement data of taxi drivers' traffic violations recorded 60 violations, such as violating traffic markings, failure to yield to pedestrians, and failure to use a safety belt. The severity of traffic violations was divided into slight infractions, medium infractions, and severe infractions. Figure 9 shows that slight and medium traffic violations account for 92.5% of the total violations. The actual distribution of traffic violations is that slight and medium traffic violations account for about 85%, and severe traffic violations account for about 15%. The data distribution studied in this study is basically in line with reality, indicating the reliability of the research data(Jin & Deng, 2017).

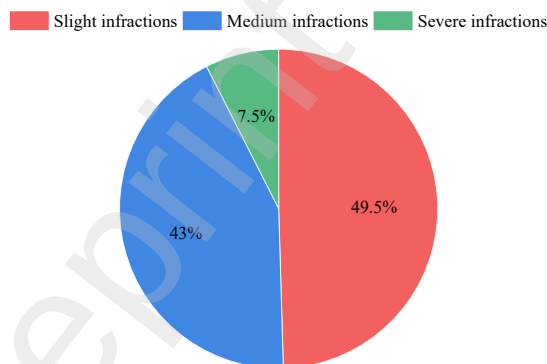


Fig. 9. Distribution of traffic violation levels.

In this study, the time periods of the traffic violation were divided into low frequency, medium frequency, and high frequency. The low frequency time period was 02:00-06:00, and the medium frequency-time was 01:00, 07:00, and 19:00: 00-23:00, high time are 0:00, 08:00-18:00. The time periods of the traffic violation were based on the period distribution of taxi drivers' traffic violations. The daily spread of the taxi drivers' traffic violations is illustrated in Figure 10

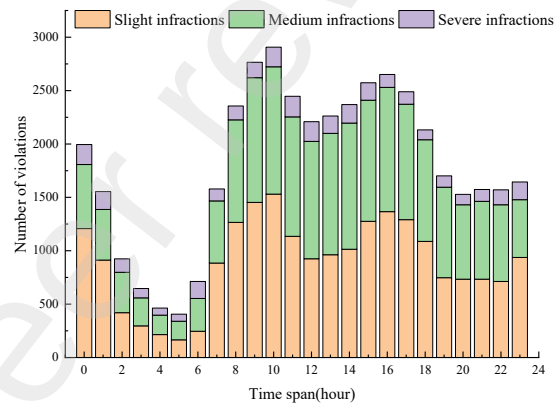


Fig. 10. Daily distribution of traffic violations.

From Figure 10, it can be seen that taxi drivers have a high proportion of traffic violations between 07:00 -18:00. The morning peak in the city is between 07:00-09:00 and the evening peak is between 17:00-19:00. Unlike the traffic congestion period, the traffic violation peak period in the morning is 9:00-10:00, and the peak period in the afternoon is 15:00-17:00. Due to the complicated road conditions in the city during peak traffic periods, the speed of the driver's vehicle slows down. In addition, during the traffic control department arrange police forces at various intersections or road sections to conduct on-site command during the morning and evening peak, drivers pay more attention to traffic laws and regulations. After the peak period, the traffic control department will reduce the police force. Drivers will be more relaxed, and traffic violations

will gradually increase. What stands out in this figure 10 is the dramatic decline in 02:00-06:00. The number of passengers and vehicles on city roads between 02:00 and 06:00 is the lowest of the day, so the number of taxi drivers' traffic violations is also the lowest.

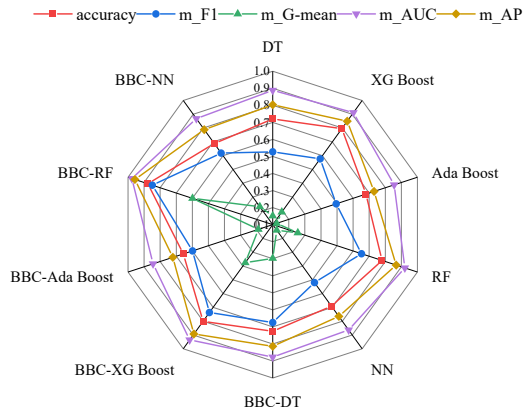


Fig. 11. Radar Maps of Different Methods. (TD = Decision Tree; RF= Random Forest; NN = Neural Networks).

A radar map (Figure 11) shows the performance of selecting imbalanced origin data or BBC processed data when different methods were utilized. Compared with machine learning algorithms such as Decision Trees, XG Boost, Ada Boost, and Neural Networks, the Random Forest model has advantages in many performance indicators. The accuracy of the random forest model is 0.877, m_F1 is 0.849, m_G-mean is 0.599, m_AUC is 0.976, and m_AP is 0.957. When the IR of the original imbalanced dataset decreased from 6.61% to 2.60%, the evaluation

indicators of the five models are improved, especially the random forest and XG Boost in the ensemble algorithm change more obviously. The m_G-mean of the random forest has increased from 0.256 to 0.599, and the m_G-mean of XG Boost has increased from 0.191 to 0.376. The m_F1 of the random forest has increased from 0.652 to 0.849, and the m_F1 of XG Boost has increased from 0.577 to 0.737. Thus, the Random Forest model using BBC with processed imbalanced data can effectively predict the severity of taxi drivers' traffic violations.

There are some unexplainable phenomena in Figure 8 in Section 5.3. For example, in the functional districts in Figure 8(a), blue represents that traffic violations occur in the central business district and old town, which has both positive and negative effects on slight infractions. Therefore, several features with interactive effects were discussed further. SHAP dependency graphs provide insight into how individual features affect the output of an RF prediction model. In addition, it can also explain the interaction between the two variables. Figure 12 and Figure 13 show the SHAP feature dependency of functional districts and location of the violation, road grade, and location of the violation, respectively. The abscissa of the SHAP dependency graph is the eigenvalue, the main ordinate is the SHAP value, and the secondary ordinate is the feature of the categorical variable.

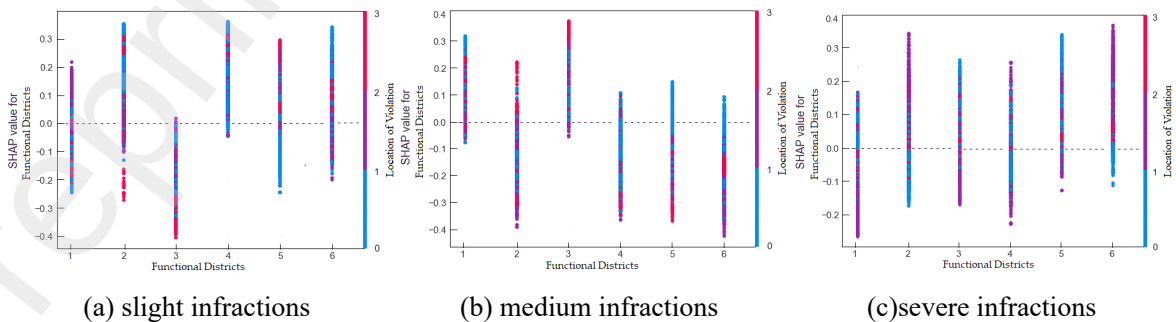


Fig. 12. Impact of functional districts and location of violation on model output.

In Figure 12(a), the SHAP value corresponding to the road intersection in the central business district is greater than zero, which indicates that slight infractions are susceptible to occur at road intersections in the central business district. The reason may be that the road system in the central business district is relatively dense, with connections between different road systems. Many road intersections make it difficult for taxi drivers to identify, resulting in slight traffic violations. The SHAP value corresponding to the road in the old town is greater than zero, indicating that the road section has a positive impact on slight infractions in the old town. In general terms, both the train and bus stations are located in the old town. Taxi driver usually waits for passengers on

the roads next to railway stations and bus stations, causing illegal parking and obstructing the passage of other vehicles and pedestrians. In Figure 12(b), the POI in the external transportation area has the greatest influence. The POI usually have more passengers distributed, and when drivers see the passengers, they will change lanes at will or directly approach the passengers from the straight road, causing the risk of changing lanes (Chin & Huang, 2009). In Figure 12(c), road intersections in old town positively affect severe infractions. The reason may be that the facilities are not perfect in the old town, and the height and angle of some signals are not standard, making taxi drivers prone to violating traffic signals.

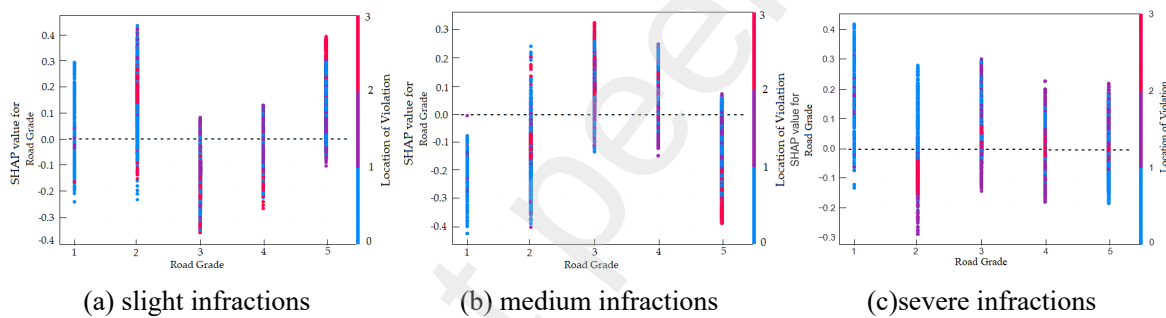


Fig. 13. Impact of functional districts and location of violation on model output.

The value of the abscissa in Figure 13 represents the category of the road grade. In Figure 13(a), the intersections of expressways are subject to slight infractions. The reason may be that traffic signs and markings are set at the intersections. In Figure 13(b), the SHAP value corresponding to the national and provincial highways is less than zero, indicating that the medium infractions on national and provincial highways negatively impact. The reason may be that medium infractions mainly occur on urban expressways, such as failure to use the safety belt as required when driving on the urban expressway or failure to follow the specified lane. Figure 13(c) shows that the SHAP values corresponding to different road grades are relatively large. This is because taxi drivers drive

as fast as possible to increase their financial income, and their driving becomes aggressive.

Based on the 43,458 electronic enforcement data on taxi drivers' traffic violations, it applied a Random Forest model to construct the severity of taxi drivers' traffic violations prediction model. An ensemble method Balanced Bagging Classifier was applied to balance the dataset to reduce the IR in the original imbalanced dataset. Using cross-validation and hyperparameter adjustment to optimize the random forest model achieved the best performance in several performance metrics. Such as accuracy, macro_F1 value, m_G-mean value, micro_AUC, and micro_PRC curve, 0.877, 0.849, 0.599, 0.976, and 0.957, respectively. The SHAP framework was employed to improve the

interpretability of the model and identify important factors affecting taxi drivers' traffic violations. The results showed that functional districts, location of the violation, and road grade were found to have a high impact on the probability of traffic violations; their mean SHAP values were 0.39, 0.36, and 0.26, respectively. The results could help traffic managers develop countermeasures to reduce traffic violations. For example, checking, rectifying, and improving signs and markings, adjusting the timing of traffic lights, adjusting the height and angle of traffic lights in the old town, formulating more flexible and accurate parking spaces on tertiary roads.

However, there still exist some limitations in the present study. This study drew data on taxi traffic violations, which lacked the information on drivers. Various studies have confirmed that driver factors can influence taxi traffic violations (Timmermans, Alhajjaseen, Reinolsmann, Nakamura, & Suzuki, 2019). Therefore, it is necessary to continue to invest in more research to explore the interactions between driver factors and typical factors.

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